National University of Computer and Emerging Sciences

Software for Mobile Devices Sessional-I Exam (SE1003)

Date: April 12th, 2025

Course Instructor(s)

Mr. Rana Waqas Ali, Mr. Usman Anwer

Total Time: **1 hour**

Total Marks: **30**

Total Questions: **2**

_____ Roll No

_____ Section

_____ Student Signature

Do not write below this line

Attempt all questions on the answer sheet

Question 1: Build a Shopping List App with following requirements. You can use java/kotlin for it.

1. MainActivity: Displays a list of items in a `RecyclerView` or `ListView`.
2. AddItemFragment: A fragment to add new items using implicit intents (camera/gallery).

Task:

1. RecyclerView/ListView Setup (7 Marks)
   - Implement `ShoppingAdapter` extends `RecyclerView.Adapter<ShoppingAdapter.ViewHolder>`.
   - Suppose you already have model class `ShoppingItem.java` having id, name, quantity and imageURI of string type.
   - Suppose you already have design file `item_shopping.xml` with `TextView` (name/quantity) and `ImageView`.
2. Fragment Implementation (7 Marks)
   - Create `AddItemFragment.java` with these hooks:
     - EditText fields for item name and quantity.
     - A button to launch camera/gallery (implicit intent).
     - A Save button to add the item to the list.
     - Override fragment lifecycle methods (`onCreateView`, `onViewCreated`). (*Keep some space empty inside these functions. You will need in upcoming parts.)*

3. Fragment Communication (9 Marks)

- Define an interface in AddItemFragment inside Fragment:

```
public interface OnItemAddedListener {
    void onItemAdded(ShoppingItem newItem);
}
```

- Make MainActivity.java , write all the required hooks and implement OnItemAddedListener and update the RecyclerView. Call "onItemAdded" function from Fragment when **Save button** will be clicked and add the ShoppingItem into the recycerview.
- On clicking **"Attach Image"**:
    - Launch an intent with Intent.ACTION_PICK (gallery)
    - Handle the result in ActivityResultLauncher and display the image in an ImageView.

Question 2 (For sections BCS 6A & 6B) (7 Marks)

You have to design a set of classes in Kotlin to store the data given in the following snippet.

order {
pizza("Margherita") {
size = "Large"
toppings("Extra Cheese", "Olives")
}
pizza() {
name= "Tikka"
size = "Medium"
toppings("Jalapenos")
}
}
	There are two classes, Order and Pizza, whereas multiple pizzas are appearing in an order. Plan appropriate data members, functions, and constructors. Order class will have a function named printSummary() to display a summary of the whole order.

Question 2. (For sections BCS 8A & BSE 8A) (7 Marks)

Suppose you have two fragments on the MainActivity i.e. ListFragment, DetailFragment and their ids are listfrag and detailfrag respectively. Your task is to hide the DetailFragment from the screen when phone is in portrait mode, while show both the fragments when phone will go into landscape mode. Write each and every step which you will do achieve the above requirements. Don't forget to write the required java code.